

Problem

The $3N+1$ sequence is generated using the following rules: If n is even, the next number is $n / 2$. If n is odd, the next number is $3 \times n + 1$. The sequence continues until it reaches 1. The cycle length of a number is the total number of terms in the sequence, including the starting number and 1.

Given two integers i and j , determine the maximum cycle length among all integers between i and j (inclusive).

If $i > j$, swap them before processing.

Input Format

A single line containing two space-separated integers:

i j

Constraints

$1 \leq i, j \leq 1,000,000$

Output Format

Print three integers:

i j maximum_cycle_length

using the original values of i and j .

Sample Input 0

1 10

Sample Output 0

1 10 20

Submissions

Leaderboard

Discussions

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Language

Python 3



```
1 # Enter your code here. Read input from STDIN. P
2 i,j=map(int,input().split())
3 a,b=i,j
4 if i>j:
5     i,j=j,i
6 m=0
7 for n in range(i,j+1):
8     x,c=n,1
9     while x>1:
10         if x%2==0:
11             x=x//2
12         else:
13             x=3*x+1
14             c+=1
15     if c>m:
16         m=c
17 print(a,b,m)
18
19
```

Line: 8 Col: 12

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☐ Test against custom input